

Complete Project Structure with Assets

```

project/
├── index.php
├── .htaccess
├── database.sql
├── README.md
├── app/
│   ├── core/
│   │   ├── Database.php
│   │   └── App.php
│   ├── controllers/
│   │   ├── LoginController.php
│   │   └── DashboardController.php
│   ├── models/
│   │   └── User.php
│   └── views/
│       ├── login/
│       │   └── index.php
│       └── dashboard/
│           └── index.php
├── config/
│   └── database.php
├── public/
│   ├── css/
│   │   ├── style.css
│   │   ├── login.css
│   │   ├── dashboard.css
│   │   └── theme.css
│   ├── js/
│   │   ├── app.js
│   │   ├── login.js
│   │   ├── dashboard.js
│   │   └── theme.js
│   ├── img/
│   │   ├── logo.png
│   │   ├── avatar-default.png
│   │   └── icons/
│   ├── fonts/
│   │   └── custom-font.woff2
│   └── vendor/
└──

```

```
└─ storage/                                ← File uploads, logs (optional)
   └─ uploads/
   └─ logs/
```

Complete Assets Management Guide

Quick Answer

Put your CSS and JS files in the `public/` folder:

```
project/
└─ public/
   └─ css/
       └─ style.css
   └─ js/
       └─ app.js
   └─ img/
       └─ logo.png
```

Three Ways to Load Assets

☒ Method 1: Using Helper Class (RECOMMENDED)

Setup:

1. Create `config/app.php` with base URL
2. Create `app/core/Helper.php` with asset methods
3. Use in views:

```
<?php require_once __DIR__ . '/../../core/Helper.php'; ?>

<!-- Load CSS -->
<link rel="stylesheet" href="<?= Helper::asset('css/style.css') ?>">

<!-- Load JS -->
<script src="<?= Helper::asset('js/app.js') ?>"></script>

<!-- Load Image -->


<!-- With cache busting -->
<link rel="stylesheet" href="<?= Helper::assetVersion('css/style.css') ?>">
```

Benefits:

- ☒ Easy to change URLs in one place
 - ☒ Works in any environment
 - ☒ Cache busting support
 - ☒ Clean, maintainable code
-

Method 2: Direct Paths (SIMPLE)

```
<!-- Absolute path from root -->
<link rel="stylesheet" href="/public/css/style.css">
<script src="/public/js/app.js"></script>

```

Benefits:

- ☒ Simple and straightforward
- ☒ No setup needed

Drawbacks:

- ✗ Harder to change if you move the project
 - ✗ No cache busting
-

Method 3: PHP Constants

In `index.php` :

```
define('BASE_URL', 'http://localhost/project');
define('ASSETS', BASE_URL . '/public');
```

In views:

```
<link rel="stylesheet" href="<?= ASSETS ?>/css/style.css">
<script src="<?= ASSETS ?>/js/app.js"></script>
```

Recommended Folder Structure

```
public/
├── css/
│   ├── style.css          ← Main styles
│   ├── login.css         ← Page-specific
│   ├── dashboard.css
│   └── components/
│       ├── buttons.css
│       ├── forms.css
│       └── cards.css
├── js/
│   ├── app.js            ← Main JavaScript
│   ├── login.js          ← Page-specific
│   ├── dashboard.js
│   └── utils.js          ← Utilities
├── img/
│   ├── logo.png
│   ├── logo-dark.png
│   ├── favicon.ico
│   └── icons/
│       ├── user.svg
│       └── dashboard.svg
├── fonts/                ← Custom fonts (optional)
│   └── custom-font.woff2
└── vendor/              ← Third-party libs (optional)
    ├── bootstrap/
    └── fontawesome/
```

Example: Complete Login Page

```
<?php require_once __DIR__ . '/../../core/Helper.php'; ?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Login</title>

    <!-- Your CSS -->
    <link rel="stylesheet" href="<?= Helper::asset('css/style.css') ?>">
    <link rel="stylesheet" href="<?= Helper::asset('css/login.css') ?>">

    <!-- OR use direct paths -->
    <link rel="stylesheet" href="/public/css/style.css">

    <!-- CDN Libraries -->
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>

    <div class="login-container">
        <!-- Logo -->
        

        <!-- Form -->
        <form action="<?= Helper::url('login/authenticate') ?>" method="POST">
            <!-- form fields -->
            </form>
        </div>

        <!-- Scripts at bottom -->
        <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js">
</script>
        <script src="<?= Helper::asset('js/app.js') ?>"></script>
        <script src="<?= Helper::asset('js/login.js') ?>"></script>
    </body>
</html>
```

.htaccess Configuration

Make sure your `.htaccess` allows access to the `public/` folder:

```
<IfModule mod_rewrite.c>
  RewriteEngine On

  # Allow direct access to public folder
  RewriteCond %{REQUEST_URI} ^/public/
  RewriteRule ^ - [L]

  # Route everything else to index.php
  RewriteCond %{REQUEST_FILENAME} !-f
  RewriteCond %{REQUEST_FILENAME} !-d
  RewriteRule ^(.*)$ index.php?url=$1 [QSA,L]
</IfModule>
```

Performance Tips

1. Load Order Matters

```
<head>
  <!-- Critical CSS first -->
  <style>
    /* Inline critical styles for above-the-fold */
  </style>

  <!-- External CSS -->
  <link rel="stylesheet" href="...">
</head>
<body>
  <!-- Content -->

  <!-- Scripts at bottom before </body> -->
  <script src="..."></script>
</body>
```

2. Minify for Production

```
# Minify CSS
npx minify public/css/style.css > public/css/style.min.css

# Minify JS
npx minify public/js/app.js > public/js/app.min.js
```

In production views:

```
<?php
$env = 'production'; // Change based on environment
$suffix = ($env === 'production') ? '.min' : '';
?>
<link rel="stylesheet" href="<?= Helper::asset("css/style{$suffix}.css") ?>">
```

3. Cache Busting

```
<!-- Automatic version based on file modification time -->
<link rel="stylesheet" href="<?= Helper::assetVersion('css/style.css') ?>">

<!-- Outputs: /public/css/style.css?v=1234567890 -->
```

4. Combine Files

production.css (combine multiple CSS files):

```
@import url('variables.css');
@import url('reset.css');
@import url('components.css');
@import url('layout.css');
```

Using CDN with Fallback

```
<!-- Load from CDN -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.min.js">
</script>

<!-- Fallback to local if CDN fails -->
<script>
if (typeof bootstrap === 'undefined') {
    document.write('<script src="<?=
Helper::asset('vendor/bootstrap/js/bootstrap.min.js') ?>"></script>');
}
</script>
```

Best Practice Summary

Aspect	Recommendation
Location	<code>public/</code> folder
Access Method	<code>Helper::asset()</code> or direct <code>/public/</code> paths
Organization	By type (<code>css/</code> , <code>js/</code> , <code>img/</code>)
CDN	Use for libraries (Bootstrap, FontAwesome)
Custom Code	Store locally in <code>public/</code>
Cache	Use versioning (<code>?v=timestamp</code>)
Production	Minify and combine files
Load Order	CSS in <code><head></code> , JS before <code></body></code>

☒ Complete Working Example

File Structure:

```
project/
├── index.php
├── config/
│   ├── app.php
│   └── database.php
├── app/
│   └── core/
│       └── Helper.php
└── public/
    ├── css/
    │   └── style.css
    └── js/
        └── app.js
```

Usage in any view:

```
<?php require_once __DIR__ . '/../..../core/Helper.php'; ?>
<link rel="stylesheet" href="<?= Helper::asset('css/style.css') ?>">
<script src="<?= Helper::asset('js/app.js') ?>"></script>
```

That's it! Simple, clean, and professional. 🚀

Troubleshooting

Assets not loading?

1. Check path:

```
<?php
echo Helper::asset('css/style.css');
// Should output: http://localhost/project/public/css/style.css
?>
```

2. Check file exists:

```
ls -la public/css/style.css
```

3. Check .htaccess allows access

4. Check browser console for 404 errors

5. Verify base_url in config/app.php

Advanced: Organizing Large Projects

For large projects, use this structure:

```
public/
├── css/
│   ├── vendor/           ← Third-party CSS
│   ├── common/          ← Shared styles
│   ├── components/      ← Reusable components
│   └── pages/            ← Page-specific styles
└── js/
    ├── vendor/
    ├── common/
    ├── components/
    └── pages/
```

This keeps your code **organized and maintainable!** 🤖